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ForceLeWM: A Modality-Robust JEPA World Model with Gated Force Fusion

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***Abstract*— In contact-rich manipulation, force/torque sensing complements vision when the end-effector occludes the workpiece. Yet force is often unavailable at deployment, and existing force-conditioned policies treat it as an always-available input. A natural remedy adds an encoded wrench to a vision world model's latent state, but this backfires. On our force-augmented PushT benchmark (PushT-FT), planning success collapses from 41% to 15% once force is missing at test time. The collapse persists under additive and concatenated fusion, and even when missing force dominates the training data. Controlled measurements point to the amplitude at which force is written into the shared latent. We propose ForceLeWM, a JEPA world model that writes force through a conservative per-dimension gate at a small learned amplitude, with a warmup alignment loss for from-scratch training. ForceLeWM restores planning to 35%, recovering about 75% of the lost success across 3 training seeds. Frozen simulation-trained checkpoints, one per task from the same recipe, run end-to-end on two real Franka assembly tasks, with 30 trials each. They succeed in 90% of gear-meshing trials and 97% of peg-insertion trials, and stay within one trial of these rates when the world model's force input is zeroed. On the real gear rig, the deployed representation decodes contact-force magnitude at $r^2{=}0.998$, against 0.693 for the vision-only model. A proprioceptive wrench observer supplies the force input from joint torques, without a force/torque sensor. When force may be absent at test time, these results favor conservative fusion that degrades gracefully rather than collapsing.**

I. Introduction

Humans resolve the final phase of contact through touch, and under digital anesthesia they drop objects far more often even with vision intact [1]. For robots, vision is least reliable exactly when precise spatial feedback matters most. The closing gripper occludes the region being aligned [2]. Parts touch before they align, and no camera reports the contact forces that then decide success [3], [4]. Perception and control latency surfaces as force overshoot or jamming [5].

Force sensing complements vision at these moments. On a simulated gear-meshing task, a vision decoder trained to recover the held gear's tooth phase stays at the $5.0°$ chance level across camera resolutions. The same decoder on the contact wrench reaches 3.3–3.5°. The wrench thus carries sub-tooth-pitch phase that the camera does not resolve at the tested resolutions (§IV-B).

Yet force is often missing at deployment. Training data is collected on sensor-equipped platforms, while the target robot mounts no force/torque sensor. Existing force-conditioned methods [6], [7], [8], [9] treat force as an always-available input. The natural remedy adds an encoded wrench to a vision world model's latent state. The model should then fall back to vision when force is absent. Instead, planning comes to depend on force, a modality shortcut [10], [11]. Once force is absent, planning collapses (§IV-C). Missing-modality methods [12], [13], [14], [15] offer inference-time remedies but assume frozen or pretrained encoders, whereas the world models studied here train from scratch.

Controlled measurements (§IV-C) point to **the amplitude at which the fusion writes force into the shared latent**. An equal-parameter concatenation variant collapses indistinguishably from addition, so the fusion operator is not the cause. Unfamiliarity with missing force is not the cause either, because the additive model still collapses when force is dropped from most training samples. A conservative gate that writes force at a small learned amplitude recovers most of the lost planning success.

We propose **ForceLeWM** (Fig. 1), which fuses force at the vision-encoder output of a from-scratch joint-embedding predictive architecture (JEPA) world model [16], [17], [18]. A conservatively initialized per-dimension gate keeps the initial write amplitude low. The vision encoder must therefore learn the spatial encoding for planning on its own while force stays at low amplitude. An alignment loss pulls the fused latent toward the vision-only latent of the same frame. It starts after a short warmup, because this vision-only anchor still drifts early in from-scratch training.

Our contributions are:

- **ForceLeWM.** A JEPA world model with conservatively gated force fusion and warmup alignment. It encodes contact force when present and degrades gracefully rather than collapsing when force is absent.
- **A modality shortcut and its mechanism.** In force-conditioned world models trained from scratch, latent planning collapses once force is absent. Controlled measurements rule out the fusion operator, unfamiliarity with missing force, and one-step dynamics error, and point to the force write amplitude.
- **Evidence across simulation and hardware.** In simulation, a three-seed PushT-FT study establishes the collapse, and ForceLeWM recovers about three quarters of the lost planning success. Train-consistent probing finds a force perception benefit on FORGE Peg, and a gear-meshing measurement shows that the wrench carries sub-tooth-pitch phase that vision does not resolve. On hardware, frozen simulation-trained checkpoints, one per task, drive gear meshing and peg insertion on physical replicas of the simulated assets, seating in 27 and 29 of thirty trials each. Zeroing the force input changes seating by at most one trial. Force comes from a proprioceptive wrench observer, not a force/torque sensor.

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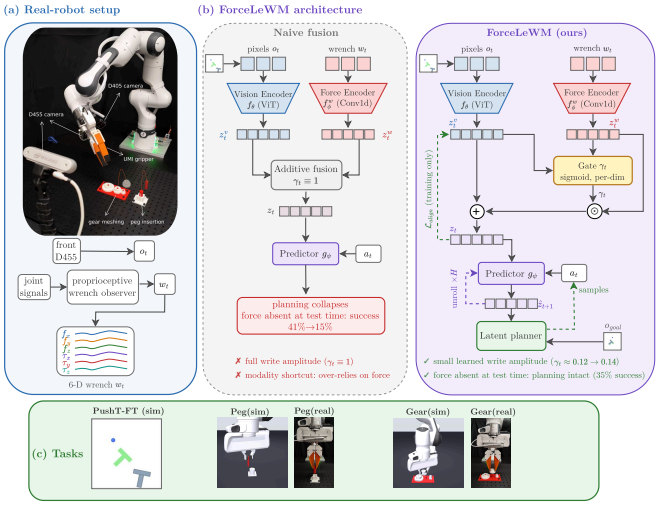


Fig. 1. **ForceLeWM overview. (a)** A world model trained entirely in simulation drives a Franka Research 3 (FR3) arm with a UMI (Universal Manipulation Interface) [19] gripper. The model's only pixel input o_t is the external front D455 view. With no force/torque sensor mounted, a learned proprioceptive wrench observer estimates the 6-D wrench w_t from joint signals. **(b)** Naive fusion (left) writes force at full amplitude ($\gamma_t \equiv 1$), and planning collapses once force is absent at test time (PushT-FT success, 41%→15%). ForceLeWM (right) writes force through a conservative per-dimension gate, $z_t = z_t^v + \gamma_t \odot z_t^w$, and a warmup alignment loss pulls z_t toward the vision-only latent. The planner scores action samples by latent distance to an encoded goal image, using receding-horizon CEM in simulation and horizon-1 greedy on the real robot. Force-absent planning recovers to 35%. **(c)** Simulated (PushT-FT, FORGE) and real tasks, shown as the model's pixel observations.

II. Related Work

a) JEPA world models: Joint-embedding predictive architectures (JEPA) [16] replace pixel reconstruction with self-supervised prediction in latent space. This line covers images [20] and video [21], [17]. Recent work extends it to zero-shot planning on pretrained visual features [22] and to visuo-tactile world modeling [23]. We build on LeWorldModel (LeWM) [18], a JEPA world model trained end-to-end from raw pixels with no pretrained encoder. We call this regime from-scratch training. Nearly all of these models take vision as their only input.

b) Noise robustness of JEPA: JEPA-style objectives implicitly suppress uninformative noise features [24]. This robustness concerns perturbations inside one modality, whereas we study a fused representation that loses an entire modality at inference.

c) Missing-modality robust learning: Methods for modalities missing at inference rely on cross-modal imputation [12], dynamic modality selection [13], masked modality projection [14], or cross-modal proxy tokens [15]. These methods assume frozen or pretrained encoders, so their alignment targets stay stable during training. ForceLeWM trains from scratch, and its alignment target still drifts early in training.

d) Contact-modality conditioning and nearest neighbors: Visual Dexterity [25], a vision-only system, hits an out-of-distribution ceiling. Most force-conditioned policies build on visual imitation learning [26] and use force as a policy input channel [6], [7], [8], rather than as part of a shared world-model latent. Closest to our force source, FILIC [27] also estimates the end-effector wrench without a force/torque sensor, with a Jacobian pseudo-inverse and a digital twin. It uses the estimate as a cross-attention query that modulates visual features and delegates force execution to a fixed impedance inner loop. Force thus never enters the propagated state. ForceLeWM instead writes force into the world-model latent, which the predictor propagates during imagination. FACTR 2 [28] learns an external torque estimator from free-motion data and uses the estimate to resample contact segments for behavior cloning. Our proprioceptive wrench observer is also calibrated on free-motion data, but its estimate enters the world-model latent and is judged by contact-event discrimination rather than regression accuracy.

At the world-model level, the nearest neighbors are visuo-tactile world models, including OmniVTA [9], Dream-TacVLA [29] on a V-JEPA backbone, force-guided TacForeSight [30], and Dream-Tac [31]. The latter three use high-dimensional tactile images, with force serving only as a conditioning signal, and assume the modality remains available at deployment. Our force signal is a low-dimensional wrench coupled to proprioception. FAWAM [32] fuses force into a world-action model in the same low-dimensional wrench regime. Its closed-loop execution presumes continuously available force feedback. The two works are complementary: FAWAM exploits force when it is present, and we address force absence. CHORD [33] uses an object-centric contact-wrench space only as training-time guidance for reinforcement learning, so its deployed policy never conditions on force. ForceLeWM keeps force as a run-time input and makes the conditioning robust to force absence.

JEPA robustness analyses cover input noise within a single modality [24]. Missing-modality methods assume pretrained encoders [15], [14]. JEPA contact modeling handles high-dimensional touch [29]. We study the fusion of a low-dimensional, proprioception-coupled force modality into the shared latent of a from-scratch JEPA world model, under force absence at planning time. This setting falls outside all three lines of work.

III. Method: ForceLeWM

A. Preliminaries and Problem Formulation

Joint-embedding predictive architectures (JEPA) [16] predict in latent space. Unlike pixel-reconstructing world models [34], [35], they skip high-frequency visual detail at the moment of contact and let the planner unroll imagined trajectories cheaply. We build on LeWM [18], a pixel JEPA world model trained from scratch. Let o_t denote the image observation at time t and a_t the action. A ViT [36] vision encoder f_θ maps the observation to a D-dimensional visual latent

$$z_t^v = f_\theta(o_t) \in \mathbb{R}^D. \tag{1}$$

The world model operates on a latent z_t, which is z_t^v in the purely visual model and the fused latent from §III-C onward.

An autoregressive predictor g_ψ predicts the next latent from the C most recent latents and actions,

$$\hat{z}_{t+1} = g_\psi\big(z_{t-C+1:t},\ a_{t-C+1:t}\big), \tag{2}$$

where $z_{t-C+1:t} = (z_{t-C+1}, \ldots, z_t)$ is the latent history and $a_{t-C+1:t}$ the corresponding actions. Actions enter through AdaLN-zero conditioning [37].

Training minimizes the one-step prediction loss in latent space,

$$\mathcal{L}_{\text{pred}} = \big\|\hat{z}_{t+1} - z_{t+1}\big\|^2, \tag{3}$$

where the target z_{t+1} is the encoder's output at the next frame. The target carries **no** stop-gradient, so encoder and predictor co-evolve. Instead of the moving-average targets of I-JEPA [20] and V-JEPA [21], [17], LeWM follows LeJEPA [38] and avoids collapse with the Sketched-Isotropic-Gaussian Regularizer (SIGReg) $\mathcal{L}_{\text{reg}}$, which constrains the latent distribution to an isotropic Gaussian. The base training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{reg}}\,\mathcal{L}_{\text{reg}}. \tag{4}$$

We keep the official LeWM defaults (D=192, C=3, λ_{reg}=0.09) without retuning and change only the force pathway (§III-B–III-D).

Planning runs model-predictive control with the cross-entropy method (CEM) [39], [40] in latent space. A goal image o_{goal} specifies the task and is encoded once as $z_{\text{goal}} = f_\theta(o_{\text{goal}})$. The planner samples action sequences, unrolls the predictor (2) autoregressively for H steps, and scores each candidate by the goal distance of its final predicted latent,

$$\text{cost} = \big\|\hat{z}_{t+H} - z_{\text{goal}}\big\|^2. \tag{5}$$

CEM iteratively refits the sampling distribution to the lowest-cost candidates, and the initial actions of the best sequence are executed with receding-horizon replanning.

a) Problem setting: As a complement to vision during contact, the model may observe a six-dimensional end-effector wrench w_t besides the image o_t. Force is abundant in training data from sensor-equipped platforms but often missing at deployment, where the target robot may mount no force/torque sensor. We write w_t=$\varnothing$ for force absence. We seek a **modality-robust** fused world model that exploits force for contact perception when present and whose planning degrades gracefully rather than collapsing when force is absent. The fused latent z_t must keep **low dependence** on its force component throughout training, because the write amplitude separates collapsing from recovering fusion (§IV-C).

B. Force Encoding and Encoder-Side Fusion

Unlike an image, the wrench $w_t \in \mathbb{R}^6$ reflects **contact dynamics**, and a single reading is ambiguous on its own. A 5 N reading may be the peak of a brief impact or the middle of a steady press. The force encoder f_ϕ^w integrates the last κ readings,

$$z_t^w = f_\phi^w\big(w_{t-\kappa+1:t}\big) \in \mathbb{R}^D, \tag{6}$$

implemented as a width-κ causal Conv1d followed by an MLP that projects to the shared latent dimension D. Temporal structure is handled at two scales: raw force over κ steps inside the force encoder, and latent dynamics over C steps inside the predictor. Both windows use three steps. The LeWM default is C=3, and we set κ equal to it without further tuning. Throughout, force is treated as a contact-event-level signal, judged by contact detection and separability rather than by force-regression accuracy. This is what lets one architecture ingest wrench sources of very different fidelity, from simulator ground truth to a proprioceptive estimate.

Conditioning the predictor g_ψ on force the way it is conditioned on actions breaks down at planning time. The planner proposes future actions, but future force readings do not yet exist. We therefore choose **encoder-side fusion**, merging the two latents at the vision encoder output into a fused latent

$$z_t = z_t^v \oplus z_t^w, \tag{7}$$

where $\oplus$ is a fusion operator. The predictor and the CEM planner both operate on this fused latent, so the unroll needs no force input beyond the current latent.

C. Conservative Element-wise Gated Fusion

We instantiate the fusion operator $\oplus$ in (7) as a conservative element-wise gate (Fig. 1b),

$$z_t = z_t^v + \gamma_t \odot z_t^w, \tag{8}$$

where the gate $\gamma_t = \sigma(\mathbf{W}[z_t^v; z_t^w] + \mathbf{b}) \in \mathbb{R}^D$, with $\mathbf{W} \in \mathbb{R}^{D\times 2D}$ and $\mathbf{b} \in \mathbb{R}^D$. Here $[\,\cdot\,;\,\cdot\,]$ denotes concatenation, $\odot$ the element-wise product, and σ the sigmoid. We initialize $\mathbf{W}$=$\mathbf{0}$ and $\mathbf{b}$=-2. The initial gate is then constant across inputs, $\gamma_t \approx \sigma(-2) \approx 0.12$, so force flows at roughly 12% amplitude per dimension. Early in training, force is nearly silent, and the vision encoder must learn the spatial encoding required for planning on its own. The gate is a D-dimensional vector, not a scalar, and the gradient raises individual dimensions only where fusion improves the prediction objective (3). The force write can thus be confined to a subspace while spatial-layout dimensions stay low, so removing force perturbs only that subspace. A scalar gate must admit or suppress force in every dimension at once.

Gated fusion also has a structural fixed point. Force absence is implemented by zeroing the force latent, $z_t^w = \mathbf{0}$, so $\gamma_t \odot z_t^w = \mathbf{0}$ and $z_t = z_t^v$ exactly, with no operator-induced distortion. The goal image carries no wrench and is encoded through the same path, so $z_{\text{goal}} = z_{\text{goal}}^v$ exactly.

D. Warmup Alignment Loss for From-Scratch Training

The conservative gate keeps the write amplitude small at initialization, and the small converged amplitude (§IV-C) emerges from the prediction loss, not from a term that enforces it. To constrain the amplitude explicitly, we add an alignment loss that pulls the fused latent toward its pure visual component,

$$\mathcal{L}_{\text{align}} = \big\|z_t - \text{sg}(z_t^v)\big\|^2 = \big\|\gamma_t \odot z_t^w\big\|^2, \tag{9}$$

where $\mathrm{sg}(\cdot)$ is the stop-gradient operator. Since $z_t - z_t^v = \gamma_t \odot z_t^w$, the two forms are identical in value, and the alignment loss is the squared **write amplitude of force into the fused latent**. It is an amplitude penalty, not representation distillation. We implement the left-hand form. Its detached anchor $\mathrm{sg}(z_t^v)$ cannot be dragged toward the fused latent, and the loss value depends on the visual latent only through the gate. Reducing the loss therefore requires shrinking the write amplitude itself.

ForceLeWM trains the vision encoder and the fusion branch jointly from scratch, so early in training f_θ has not converged and the anchor drifts. Applied from epoch one, alignment pulls the fused latent toward a moving target, and planning regresses below gating alone (§IV-D). We therefore use a two-stage schedule: alignment stays off for the initial T_w epochs and activates once the vision encoder has stabilized. We attribute this warmup requirement to the from-scratch regime. The full training objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{reg}}\,\mathcal{L}_{\text{reg}} + \mathbf{1}[\,e \geq T_w\,]\,\lambda_{\text{align}}\,\mathcal{L}_{\text{align}}, \tag{10}$$

where e is the current epoch and $\mathbf{1}[\cdot]$ the indicator function. We use λ_{align}=0.1 and T_w=5 of 10 training epochs in all experiments, set once and not tuned.

a) Force dropout during training: We also randomly zero the force latent z_t^w with probability p_{drop}=0.3 (per sample, over the whole window), creating force-absent training samples with $z_t = z_t^v$. The gate initialization, the alignment loss, and force dropout constrain the fused latent's force dependence on three complementary fronts: architecture, loss, and data.

E. Proprioceptive Wrench Observer

The target robot mounts no force/torque sensor, so the wrench w_t must come from the arm's own joint signals. The arm computes an external joint torque as the residual of the measured torque τ_t against its rigid-body model,

$$\tau_t^{\text{ext}} = \tau_t - \left[M(q_t)\ddot{q}_t + C(q_t,\dot{q}_t)\dot{q}_t + g(q_t)\right], \tag{11}$$

with joint positions q_t, mass matrix M, Coriolis matrix C, and gravity g. It maps this residual to an end-effector estimate $w_t^{\text{raw}} = (J(q_t)^\top)^+ \tau_t^{\text{ext}}$ through the Jacobian pseudo-inverse [41]. Any mismatch between the model and the physical arm appears in w_t^{raw} as a spurious wrench. This bias depends on the arm state. It drifts with pose and grows with velocity and acceleration, so a constant tare cannot remove it.

The observer learns the bias from contact-free motion, where the true external wrench is zero and every nonzero reading is model mismatch:

$$\hat{w}_t = w_t^{\text{raw}} - b_\phi(x_t), \qquad b_\phi(x) = Wx + c. \tag{12}$$

The feature $x_t \in \mathbb{R}^{44}$ stacks the end-effector pose (16), joint positions, velocities, measured torques, and external-torque residuals (7 each). Ridge regression fits b_ϕ to w_t^{raw} on contact-free frames, with the penalty selected by five-fold cross-validation. At deployment the residual $\hat{w}_t$ retains the contact wrench, and the world model receives $w_t = \hat{w}_t$. The observer is a data-driven instance of the generalized momentum observer [42]: subtracting a model of the robot's own expected dynamics leaves the external force. It targets contact-event detection rather than calibrated force regression (§III-B).

IV. Experiments

A. Experimental Setup

a) Evaluation platforms: We evaluate on four platforms. PushT-FT, the controlled planning testbed, is a force-augmented 2D planar pushing benchmark [26]. Its contact wrench is recovered exactly from engine collision impulses. The public REASSEMBLE [43] dataset of real Franka assembly with six-axis force/torque sensing spans five task families and audits the breadth of the perception benefit. Two FORGE [44] simulation tasks, gear meshing (*Gear*) and peg insertion (*Peg*), form the deployment family. A real Franka Research 3 (FR3) with a UMI [19] gripper runs both tasks on replicas printed from the FORGE assets (§IV-E).

b) Compared models: The vision-only **baseline** is LeWM (§III-A). Two force-fused models extend it with the same causal Conv1d force encoder and differ only in the fusion, the controlled variable. **Additive** fusion writes force into the shared latent at full amplitude (naive fusion). Our **gated+align** model, **ForceLeWM**, uses conservative gated fusion with warmup alignment. LeWM reaches 94% on the original PushT with its official recipe of 100 training epochs [18]. All arms here share the shorter PushT-FT training and planning protocol below, so the comparison between fusion designs is controlled. Absolute success rates are not directly comparable to planners on pretrained visual features such as DINO-WM [22].

c) Metrics and seeds: Linear probes on the frozen encoder's latent measure perception. P1 regresses the six-dimensional contact wrench (r^2). P2 classifies contact against no-contact frames (balanced accuracy, chance 0.5). P3a regresses object or end-effector position (r^2), the visual-spatial information planning relies on. Planning is measured by latent-space CEM [39], [40] success rate on PushT-FT under force absence, over fifty episodes per seed. The CEM samples 300 action sequences per step over horizon 5, with 30 refinement iterations, top-30 elites, and receding-horizon replanning, on models trained for 10 epochs on 2,000 episodes. Under fixed code and seed the evaluation is deterministic, so repeated runs of one checkpoint reproduce identical rates. Results are means and standard deviations over three random seeds unless noted, with p-values from paired t-tests over seeds. ForceLeWM results on FORGE perception and the real robot use the deployed checkpoint of each task.

d) Role of each platform: PushT-FT is the planning testbed. FORGE and REASSEMBLE test perception (§IV-B), and the real robot tests deployment (§IV-E). On hardware, the world model selects descent and yaw with a horizon-1 greedy planner over the same latent goal cost.

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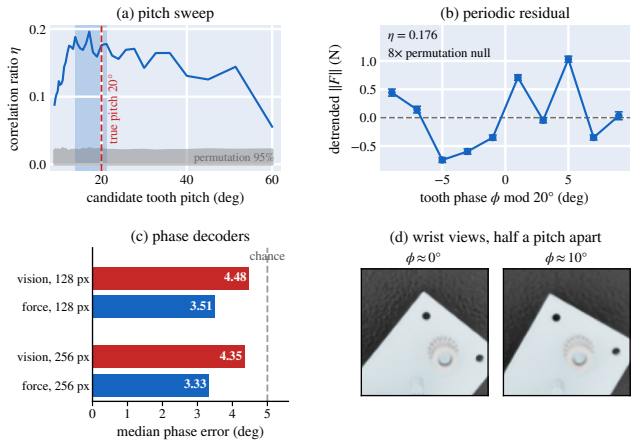


Fig. 2. Force carries gear tooth phase (FORGE gear meshing, 37,238 meshing-window frames). (a) Correlation ratio η between tooth phase and detrended $\|F\|$ across candidate pitches. The grey band is the permutation 95th percentile, and the shaded plateau covers the true 20° pitch. (b) Detrended $\|F\|$ against phase modulo 20° (mean and standard error). (c) Median phase error of decoders trained only on phase. Chance is 5.0°. (d) Pose-matched wrist views half a tooth pitch apart.

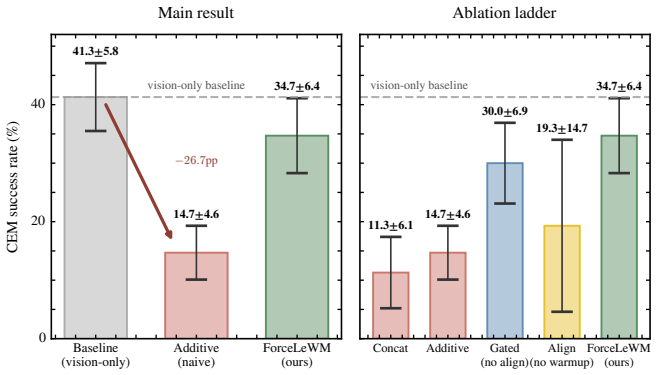


Fig. 3. **PushT-FT controlled study (force-absent, 3 seeds × 50 episodes, error bars ±1 s.d. over seeds).** Left: additive fusion collapses CEM planning (−26.7pp), and ForceLeWM recovers about 75% of the loss. Right: concatenation collapses like addition, the gate alone recovers most of the loss, and alignment without warmup regresses.

B. Perception under Train-Consistent Probing

a) Force resolves tooth phase that vision does not: On FORGE gear meshing, we test whether the wrench carries the held gear's tooth phase, its yaw relative to the gear base modulo the 20° tooth pitch. The data are 250 scripted-expert episodes. Within the meshing window, the detrended force magnitude varies with tooth phase (Fig. 2a,b). Its correlation ratio is η=0.176, about eight times the permutation bound. Image decoders trained only to recover the phase stay near the 5.0° chance level at both tested resolutions, while a wrench MLP reaches 3.3–3.5° (Fig. 2c). A tooth spans about two pixels in these views (Fig. 2d). The wrench therefore carries sub-tooth-pitch phase that the cameras do not resolve.

b) Protocol: The probes of §IV-A are fit on REASSEMBLE and on FORGE (248 Gear and 478 Peg episodes, 1.5k probe frames per task). Probes z-score the force input to the encoder's training convention.

c) Breadth on public real data: Across the REASSEMBLE families, train-consistent probes change contact discrimination by −2.3 to +3.8 pp, excluding the degenerate Pick control.

d) Perception benefit: On FORGE *Peg*, force conditioning raises contact discrimination from 0.694 ± 0.030 (vision-only, three seeds) to 0.861 (deployed ForceLeWM). Wrench decoding rises from near chance, and position decoding does not degrade. On simulated *Gear*, gains stay within seed variance, and wrench decoding on task-rollout frames stays near chance. On the real gear rig, a fixed-pose probe of the same checkpoint decodes contact-force magnitude (§IV-E(b)).

C. The Modality Shortcut: Naive Fusion Collapses Without Force

a) Naive fusion collapses planning: Additive fusion drops force-absent CEM success on PushT-FT from the baseline's 41.3 to 14.7 percent (−26.7pp, p<0.001; Fig. 3).

b) Constraining the mechanism by exclusion: Four measurements rule out the intuitive explanations. (i) *Not the operator.* Equal-parameter concatenation collapses indistinguishably from addition (p=0.62). (ii) *Not input novelty.* Per-sample force dropout zeroes the whole force embedding in training, exactly as at evaluation, yet the additive model still collapses at every tested dropout rate (§IV-F). (iii) *Small latent shift, intact one-step prediction.* Removing force shifts the fused latent by about five percent of the embedding scale for additive fusion and about two percent for ForceLeWM. One-step prediction error is essentially unchanged in both. (iv) *Not a damaged representation.* Under train-consistent normalization, additive probes match the baseline (block-position r^2 0.97 for both). The collapse therefore arises downstream of one-step prediction, where small latent shifts can compound over the multi-step rollout and the goal-distance cost. Write amplitude is the design variable that differs between collapsing and recovering fusion. The conservative gate starts near $\sigma(-2)\approx 0.12$, converges to a measured mean of 0.133, and lands in the recovering regime (§IV-D).

c) Force perception under the conservative gate: In visually rich domains such as FORGE and the real robot, the gate stays near its initialization (mean 0.10–0.12). At this amplitude, ForceLeWM's FORGE Peg contact discrimination reaches 0.861, against 0.781 ± 0.021 for additive fusion (§IV-B). The same checkpoints drive the real deployment (§IV-E). The perception gain comes from the live force input and is not distilled into the vision encoder. Without the encoder's force input, additive contact discrimination on FORGE returns to the baseline level (Peg 0.698 vs. 0.694 ± 0.030). Position probes match with force present or absent (within 0.01 r^2).

D. Gated Fusion Preserves Planning

a) ForceLeWM sustains planning: Under force absence, ForceLeWM reaches 34.7 percent CEM success, recovering about three quarters of the lost success (+20.0pp over additive, p=0.003; Fig. 3). Gating alone drives most

of the recovery, lifting success from 14.7 to 30.0 ± 6.9 ($+15.3$pp, paired p=0.049, three seeds). Warmup alignment adds a further $+4.7$pp, which is not significant on its own. Without warmup, alignment chases an unstable visual anchor and falls below gating alone.

b) The gate learns graded damping: The frame-level gate mean is *inversely* modulated by force magnitude on PushT-FT, FORGE Peg and Gear, and REASSEMBLE (episode-paired $p<10^{-8}$ on every platform). Damping is deepest on PushT-FT, where additive fusion collapses, with the gate mean dropping from 0.150 off contact to 0.116 on contact. It is milder on FORGE and smallest on REASSEMBLE. The inverse direction persists without the alignment loss and across dropout strengths, so it comes from the base training objective. The gate thus writes force at lower amplitude when contact force is large.

E. Real-Robot Deployment

a) Deployability: One training recipe, applied per task in simulation, yields frozen checkpoints that drive two physically distinct assembly tasks, gear meshing and peg insertion, on printed replicas of simulator assets without real-robot training or fine-tuning (Fig. 4). The gear checkpoint reaches a seated end state absent from its training data. The scripted collector never seats the mesh, and the goal comes from encoding a meshed pose staged once at deployment. Switching tasks swaps the checkpoint and three deployment-layer constants (simulation z-scoring statistics, staged goal embedding, kinesthetic anchor). The rest of the stack, including the three-stage execution, is shared.

The third-person camera is placed by hand to approximate the simulated view, with only exposure and white balance fixed. Absolute task-frame targets with per-step clamps at their nominal values reproduce the simulator's action semantics exactly, verified by a fourteen-case unit test. No timing compensation bridges the 15 Hz training rate and the 26 Hz deployed loop. The round 8 mm peg has no angular phase. It stays at its calibrated yaw and seats by compliant lead-in capture. A horizon-1 greedy planner (§IV-A) commands descent and yaw under Cartesian impedance control [45] via `libfranka` [41], with lateral targets locked to the calibrated anchors.

The proprioceptive wrench observer (§III-E) supplies the world model's force input at every control step, without a force/torque sensor. Evaluation spans 4,474 frames from three peg-socket contact runs and 13,620 frames of contact-free motion. The correction raises frame-level contact AUC from 0.52 to 1.000. It lowers the minimum detectable contact from 3.6 to 0.67 N.

Gear meshing seats in 27 of 30 trials with precision 0.13 ± 0.07 mm (Table I, Fig. 5a). Peg insertion seats in 29 of 30, with seated and rim-jammed outcomes about 24 mm apart in final height. On the peg, the force-capped model-driven descent operates at 4.0 ± 0.2 N, against 14 N uncapped. The cap keeps the load below where the compliant fingers release a plain cylinder, so seated depth repeats to 0.03 mm.

TABLE I

Real-robot evaluation of the frozen per-task gated+align checkpoints (154 trials). Seating and peak force are mean and standard deviation over seated trials, and peak is the maximum force during press-in. Gear seating is height above the bolt (success within 0.5 mm). Peg seating is depth relative to the kinesthetic reference (success requires seating, not rim-jamming). In (c), the baselines are the arm's raw wrench estimate and the vision-only model.

Task	Force input	Success	Seating (mm)	Peak (N)
(a) Deployment, 30 trials per row				
Gear	on	27/30	0.13 ± 0.07	7.5 ± 0.1
Gear	zeroed	26/30	0.11 ± 0.07	7.5 ± 0.06
Peg	on	29/30	-1.84 ± 0.03	6.4 ± 0.3
Peg	zeroed	28/30	-1.80 ± 0.02	6.2 ± 0.2
(b) Gear robustness, 34 trials, peak force $\leq$*8.05 N*				
Unperturbed control		4/4		
Start pose ($\pm 30^\circ$ yaw, $\pm 2/\pm 4$ mm)		12/12		
Front camera occluded		5/6		
Calibration offset 2–4 mm		3/6		
Calibration offset 5–6 mm		0/6		
(c) Force sensing on hardware		Baseline	Ours	
Contact AUC (observer)		0.52	1.000	
Min. detectable contact (N)		3.6	0.67	
Force-decoding r^2 (probe)		0.693	0.998	

b) Hardware probe: A decorrelating protocol probes force decoding on the real gear rig: at a fixed end-effector pose, randomized force targets of 1.3–6.1 N are commanded in non-monotonic order. Cross-validation keeps each continuous command burst within one fold, against near-duplicate leakage. The deployed model's representation decodes contact-force magnitude at r^2=0.998, against 0.693 for the vision-only baseline (Fig. 5b). Probe inputs follow the training normalization.

c) Force-absent deployment: With the world model's force input zeroed, 26 of 30 gears and 28 of 30 pegs seat, within one trial of force-on (Table I; dashed lines in Fig. 4). Seated depth matches force-on to within 0.04 mm. Both conditions share the calibrated lateral anchor and the force-controlled press-in, so this test measures the deployed stack's robustness to force absence rather than the gate's contribution. With the front camera fully occluded, gear seating completes in five of six trials.

d) Gear failure boundary: Start-pose perturbations ($\pm 30^\circ$ yaw, ± 2 and ± 4 mm translation) are absorbed in all twelve trials, as the calibration-locked lateral target predicts (see (a)). A four-trial unperturbed control seats 4/4. Injected calibration offsets, which displace the target itself, fail sporadically at 2, 3 and 4 mm, where 1 of 2 trials seats at each. From about 5 mm they fail consistently, and all six trials jam. Jammed gears sit against the base at heights up to about 20 mm, as does the single occlusion failure (Fig. 5c). Jams end by descent stall or the 8 N contact-force stop. Peak force never exceeds 8.05 N across the 34 robustness trials. The 15 N hard safety limit is never approached, so the failures are geometric.

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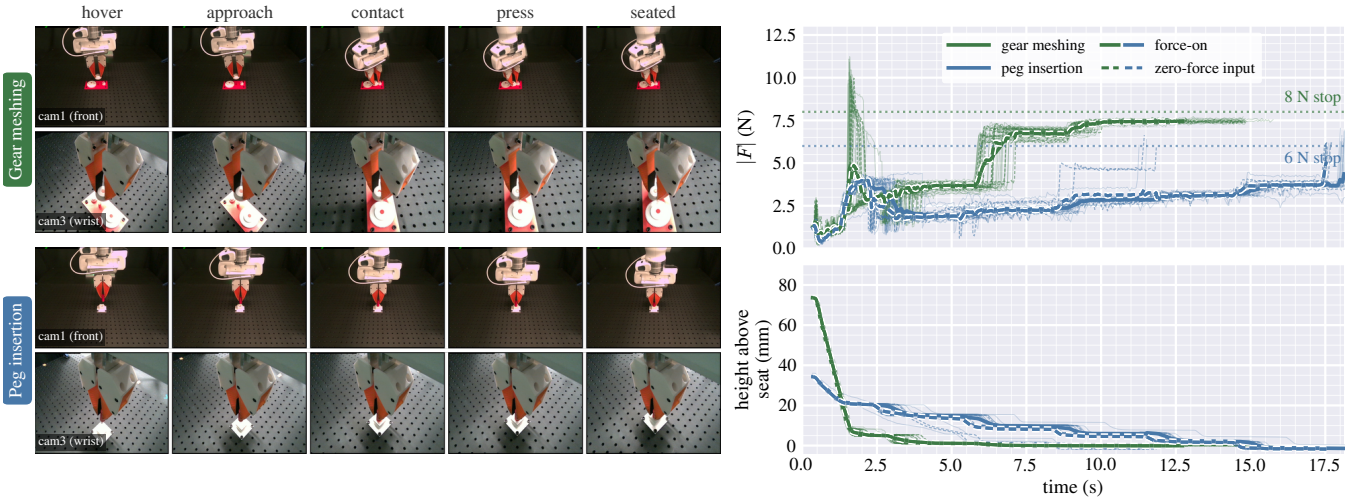


Fig. 4. One recipe, two real-robot assembly tasks. Strips show one successful rollout per task, chosen for visibility, from the front camera (cam1, model input) and wrist camera (cam3, monitoring only). The peg strip was recorded during protocol development. Right: built-in wrench-estimate magnitude $|F|$ and height above the seated pose. Solid lines are force-on trials (thirty gear, twenty peg), dashed lines are trials with the world model's force input zeroed (ten per task), and bold lines are the means of each condition. Dotted lines mark the press-in thresholds (8 N gear, 6 N peg). All peg trials cross their threshold in brief terminal spikes that the means average out, and the gear press-in ends by descent stall at 7.5 N.

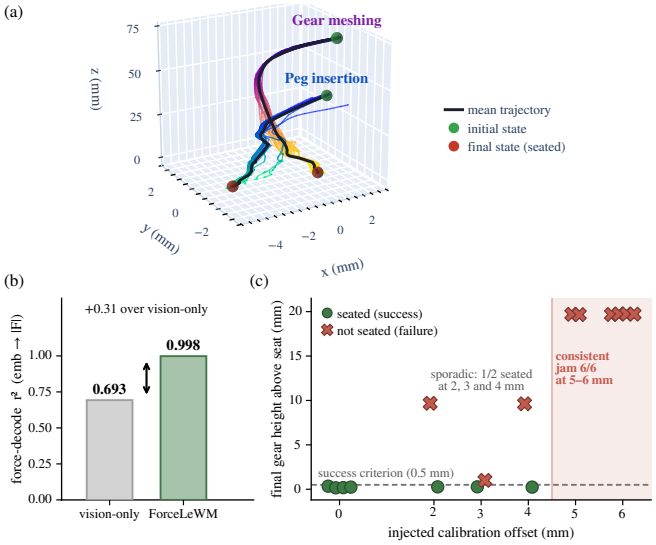


Fig. 5. Real-robot rollouts, probe, and failure boundary. (a) End-effector trajectories from hover to goal, shown for thirty gear and twenty peg force-on trials, centered on each task's seated pose, with per-task means in black. (b) On the real gear rig, the deployed gated+align representation decodes contact-force magnitude at r^2=0.998 versus 0.693 for the vision-only baseline (chance r^2=0). (c) Gear-meshing failure boundary: injected calibration offsets seat 1 of 2 trials at 2, 3 and 4 mm and jam in all six trials from 5 mm, at heights up to about 20 mm.

F. Ablations and Limitations

a) Ablations on PushT-FT: (i) *Architecture, not operator.* Both full-amplitude paths we tested collapse, with equal-parameter concatenation indistinguishable from addition (p=0.62). (ii) *Architecture, not dropout rate.* In a single-seed robustness sweep over force-dropout rates $\{0.1, 0.3, 0.5, 0.7\}$, every additive point collapses and every gated point recovers, with no overlap. (iii) *Warmup necessity.* Alignment from the start of training chases an unstable visual anchor and reaches 19.3 ± 14.7, below gating alone, whereas two-stage warmup gives the best result (Fig. 3).

b) Limitations: (i) Under force absence, ForceLeWM's CEM success (34.7%) stays below the vision-only baseline (41.3%). The gap is 6.7pp (p=0.010). Prior force-conditioned policies report gains from force [6], [7], [8], [9]. This paper addresses how to fuse force without making planning depend on its availability. (ii) On PushT-FT, supplying the true wrench at every evaluation step leaves success unchanged, so this benchmark measures robustness to force absence rather than benefit from force. (iii) The hardware study runs under normal vision. Measuring the value of force under controlled visual degradation is future work. (iv) The planning-collapse evidence comes from PushT-FT, where from-scratch CEM succeeds often enough to separate fusion designs. On force-augmented Cube, FORGE Peg with synthesized goals, and FetchPush, the vision-only baseline itself plans too rarely for that comparison (at most 6 of 144 episodes succeed on FORGE Peg). (v) The observer's contact-detection accuracy is characterized in one calibration session on one contact geometry. Calibrated force-magnitude tracking lies outside its design goal (§III-E).

V. Conclusion

We presented **ForceLeWM**, a JEPA world model with gated force fusion and warmup alignment. Its conservative gate keeps the force write amplitude small throughout training and recovers most of the collapsed planning performance across seeds (§IV-C–IV-D). On FORGE Peg, force conditioning makes the contact wrench linearly decodable without degrading vision (§IV-B). On the real gear rig, the deployed representation decodes contact-force magnitude. Frozen simulation-trained checkpoints, one per task, drive real gear meshing and peg insertion, seating in 27 and 29 of

thirty trials each. Zeroing their force input changes seating by at most one trial.

The contribution is a fusion world model that degrades gracefully when force is absent, paired with a proprioceptive wrench observer that recovers its force input from the arm's own joint signals, at a frame-level contact AUC of 1.000 without a force/torque sensor.

a) Future work: Closed-loop force planning and a hardware study under controlled visual degradation are the next steps (§IV-F).

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